Students's Name: ……

Signature: ……Class & Stream: ……..……..

P510/1
PHYSICS
Paper 1
(Theory)
2 hours 15 minutes

UGANDA STEM TEACHERS NETWORK (USTEN)

Uganda Advanced Certificate of Education

PHYSICS REVISION ASSESSMENT SET TWO

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of **THREE compulsory** items.
- Responses to **each** item should start on a **fresh page**.
- Silent non-programmable scientific calculators may be used.

ITEM 1

A paper manufacturing company is planning to install an automatic thickness-control system. The proposed system uses radioactive strontium-90, $^{90}_{38}Sr$, as a beta-particle source. Beta radiation passing through the moving paper is detected on the opposite side, and changes in the detected intensity are used to monitor the thickness of the paper.

Before purchasing the equipment, the company's technical department wants to establish whether the proposed radioactive source has an acceptable activity and whether the energy associated with its decay is sufficient for the intended operation.

Strontium-90 undergoes beta decay to form yttrium-90. The company has also proposed using an electron-beam X-ray inspection system to examine defects in finished products.

An Advanced Level Physics learner has been asked to analyse the information and explain the operation of the proposed radiation-producing device.

Support material

- Mass of $^{90}_{38}Sr$ = 89.90774 U
- Mass of $^{90}_{39}Y$ = 89.90714 U
- Half-life of $^{90}_{38}Sr$ = 28.79 years
- Mass of source = 15 g
- Molar mass of ^{90}Sr = 90 g mol^{-1}
- Avogadro constant = 6.02×10^{23} mol^{-1}
- 1 U = 931 MeV/c^2
- Maximum acceptable activity = 8.0×10^{13} Bq
- Minimum required decay energy = 0.50 MeV

Task

You have been selected to assist the technical department.

a) Analyse the supplied information to establish whether the proposed strontium-90 source is appropriate for the thickness-control system.
b) The company intends to use an electron-beam X-ray system to inspect manufactured materials. Explain the structure and working principle of the X-ray tube, including the roles of the electron source, accelerating potential, target and vacuum.

ITEM 2

A secondary school constructed a new computer laboratory with a large glazed wall to provide natural illumination. Shortly after the laboratory was opened, learners reported that the room became excessively warm during the afternoon.

The supplier claimed that the double-glazed panels had been designed to maintain the laboratory at or below 24°C when the external surface temperature of the glass reached 38°C.

Each panel consists of two glass sheets, each 6 mm thick, separated by a 15 mm air gap. The total area of the glazed wall is 18 m^2. The wall receives solar radiation of intensity 1100 W m^{-2}, of which 55% is absorbed by the glazing system.

The supplier quoted the thermal conductivity of the glass as 1.00 W m^{-1} K^{-1}, while that of the trapped air was given as 0.026 W m^{-1} K^{-1}.
The Physics teacher suspected that the quoted value for the glass and the claimed indoor temperature needed verification. A Senior Six learner was given a rectangular sample of the same glass. The sample had an area of 0.050 m^2 and thickness 25 mm. When a steady temperature difference of 12 K was maintained across the sample, the rate of heat transfer was found to be 22.8 W.
The school plans to reduce the heating effect by applying a suitable coating to the outside surface of the glazed wall. It also wants to install a device at the laboratory entrance that can automatically detect a person entering the room after working hours.

Support Material

- Thermal conductivity of air = 0.026 W m^{-1} K^{-1}
- Quoted thermal conductivity of glass = 1.00 W m^{-1} K^{-1}
- Solar radiation intensity = 1100 W m^{-2}
- Fraction absorbed by glazing = 55%
- Area of glazed wall = 18 m^2

Task

You have been selected by the Physics teacher.

a) Using the information provided, analyse the situation and determine whether the claims made by the supplier about the thermal performance of the glazed wall are reasonable. Include an appropriate calculation using the glass sample to verify its thermal conductivity and use the result to assess the heat transfer through the window system.
b) Explain how the external surface of the glazing could be modified to reduce absorption of solar radiation and describe the operation of a photoelectric beam security system that could be installed at the laboratory entrance.

ITEM 3

A telecommunications company is preparing to launch a communications satellite to improve connectivity in remote parts of Uganda. The satellite has a mass of 500 kg and is to be placed in a circular orbit around the Earth.
The engineering team has specified that the satellite should be placed in an orbit where its total mechanical energy is -1.0×10^{10} J. Before the launch is authorised, the engineers must determine the corresponding orbital height and consider how orbital energy is related to the satellite's position around the Earth.
At the same time, an emergency technician at the launch facility has been instructed to transport an important component to the satellite-control station. He uses a utility vehicle of mass 2200 kg on a straight road. The engine provides a driving force of 1100 N and develops a power of 24.2 kW.
The vehicle starts from rest and accelerates uniformly until it attains the speed corresponding to the engine power and driving force. It travels 1.0 km during the acceleration stage. The remaining 0.70 km is intended for braking.
The coefficient of friction between the vehicle's tyres and the road is 0.025.
The engineering team is also interested in developing a small locally assembled unmanned aerial vehicle to carry a communication camera above the launch facility.

Support Material

- Mass of satellite = 500 kg
- Mechanical energy = -1.0×10^{10} J
- Radius of Earth, RE = 6.4×10^{6} m
- g = 9.81 m s^{-2}
- Vehicle mass = 2200 kg
- Driving force = 1100 N
- Engine power = 24.2 kW
- Coefficient of friction = 0.025
- Acceleration distance = 1.0 km
- Braking distance = 0.70 km
- Time available = 5 minutes

Task

You have been selected to assist the engineering team.

a) Analyse the supplied data and provide the information required before the vehicle and satellite operations are authorised.

b) Explain how a small multi-propeller unmanned aerial vehicle can generate sufficient lift to carry a camera. Describe the role of the motors, propellers, frame, power supply and control system.

END

KATO IVAN WUUNA
+256 788 463 703
DOWNLOAD MORE RESOURCES FROM WUNNA DIGITAL LIBRARY